

Ex.no:2	EXPLORATORY DATA ANALYSIS (EDA) – DATA IMPORT AND EXPORT

Aim:

To inspect dataset features, detect null values, and export data into CSV, Excel, and SQLite database formats.

Algorithm:

1. Import pandas, matplotlib, and other required libraries.
2. Upload and read the Daily Google Search Trends US dataset using `pd.read_csv()`.
3. Inspect the dataset using `.head()`, `.tail()`, `.info()`, `.describe()`, `.shape`, `.columns`, and `.isnull().sum()`.
4. Perform basic data selection and filtering by selecting single and multiple columns and applying suitable conditions.
5. Analyze categorical and numerical features by checking value counts, duplicate values, and numerical column distributions.
6. Visualize the numerical data using a histogram and display the results for exploratory data analysis.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import glob
import os
print("Libraries imported successfully.")
!pip -q install kagglehub
import kagglehub
dataset_path = kagglehub.dataset_download("adrianjuliusaluoch/daily-google-search-trends-us")
csv_files = glob.glob(os.path.join(dataset_path, "**", "*.csv"),
recursive=True)
csv_path = csv_files[0]
df = pd.read_csv(csv_path)
print("DataFrame loaded successfully.")
print("Number of records:", len(df))
print("Number of columns:", len(df.columns))
print("First five records:")
display(df.head())
```

```
print("Last five records:")
display(df.tail())
print("Complete DataFrame:")
display(df)
print("Shape of DataFrame:", df.shape)
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
print("Column names:")
for i, col in enumerate(df.columns, start=1):
    print(i, "->", col)
print("Data types of each column:")
print(df.dtypes)
print("Dataset Information:")
df.info()
print("Statistical description:")
display(df.describe(include="all").T)
series1 = df.iloc[:, 0]
print("Series:")
display(series1.head())
print("Type:", type(series1))
column_name = df.columns[0]
single_column = df[column_name]
print("Selected column:", column_name)
display(single_column.head())
selected_columns = df.columns[:3].tolist()
print("Selected columns:", selected_columns)
display(df[selected_columns].head())
first_col = df.columns[0]
if pd.api.types.is_numeric_dtype(df[first_col]):
    condition = df[first_col] >= df[first_col].median()
else:
    condition = df[first_col].notna()
filtered_df = df[condition]
print("Filtering condition applied on:", first_col)
print("Filtered rows:", len(filtered_df))
display(filtered_df.head())
missing = df.isnull().sum()
```

```

print("Missing values in each column:")
display(missing)
print("Total missing values:", missing.sum())
duplicates = df.duplicated().sum()
print("Number of duplicate records:", duplicates)
text_columns = df.select_dtypes(include="object").columns
if len(text_columns) > 0:
    value_col = text_columns[0]
    print("Value frequency for:", value_col)
    display(df[value_col].value_counts(dropna=False).head(10))
else:
    print("No text/categorical column was found.")
numeric_columns = df.select_dtypes(include=np.number).columns
if len(numeric_columns) > 0:
    num_col = numeric_columns[0]
    plt.figure(figsize=(10, 5))
    plt.hist(df[num_col].dropna(), bins=20)
    plt.title("Distribution of " + num_col)
    plt.xlabel(num_col)
    plt.ylabel("Frequency")
    plt.show()
else:
    print("No numeric column was found for histogram visualization.")

```

Output:

```

Downloading to /root/.cache/kagglehub/datasets/adrianjuliusaluocho/daily-google-search-trends-us/194.archive...
100%|██████████| 935k/935k [00:00<00:00, 80.4MB/s]
Extracting files...
Kaggle dataset downloaded successfully.
Dataset folder: /root/.cache/kagglehub/datasets/adrianjuliusaluocho/daily-google-search-trends-us/versions/194
CSV files found: ['/root/.cache/kagglehub/datasets/adrianjuliusaluocho/daily-google-search-trends-us/versions/194/trending_searches_in_us.csv']
DataFrame loaded successfully.
Number of records: 31804
Number of columns: 8

```

Expected dataset size from the Kaggle listing: 31,804 rows.
The exact column count and file contents are read directly from Kaggle at runtime.

First five records:

	query	start_date	end_date	active	search_volume	increase_percentage	categories	trend_
0	shedeur sanders	2026-03-30 18:40:00+00:00	NaN	True	200	50	Sports	
1	whoopi goldberg	2026-03-30 18:40:00+00:00	NaN	True	5000	50	Politics	
2	mo williams	2026-03-30 18:00:00+00:00	NaN	True	5000	300	Sports	ma baske
3	sloane stephens	2026-03-30 17:50:00+00:00	NaN	True	1000	200	Sports, Health	
4	canceled tv shows 2026	2026-03-30 17:50:00+00:00	NaN	True	5000	200	Entertainment	r cancel

Last five records:

	query	start_date	end_date	active	search_volume	increase_percentage	catego
31799	snow squall	2026-01-19 16:00:00+00:00	2026-01-19 17:10:00+00:00	False	2000	700	O
31800	bernice king	2026-01-19 15:50:00+00:00	2026-01-19 17:00:00+00:00	False	500	100	Pol
31801	ana	2026-01-19 15:50:00+00:00	2026-01-19 16:40:00+00:00	False	500	50	Travel Transportat Business Finan
31802	peso pluma	2026-01-19 15:50:00+00:00	2026-01-20 08:20:00+00:00	False	10000	100	Entertainm
31803	tessa thompson	2026-01-19 15:20:00+00:00	2026-01-19 16:30:00+00:00	False	1000	400	Entertainm Beauty Fashi

Complete DataFrame:

	query	start_date	end_date	active	search_volume	increase_percentage	categories	trend_breakdown
0	shedeur sanders	2026-03-30 18:40:00+00:00	NaN	True	200	50	Sports	NaN
1	whoopi goldberg	2026-03-30 18:40:00+00:00	NaN	True	5000	50	Politics	america first award, mike johnson
2	mo williams	2026-03-30 18:00:00+00:00	NaN	True	5000	300	Sports	mason williams basketball, mason williams
3	sloane stephens	2026-03-30 17:50:00+00:00	NaN	True	1000	200	Sports, Health	NaN
4	canceled tv shows 2026	2026-03-30 17:50:00+00:00	NaN	True	5000	200	Entertainment	renewed and cancelled tv shows 2026
...	...	...	...	...	...	...	...	...
31799	snow squall	2026-01-19 16:00:00+00:00	2026-01-19 17:10:00+00:00	False	2000	700	Other	NaN
31800	bernice king	2026-01-19 15:50:00+00:00	2026-01-19 17:00:00+00:00	False	500	100	Politics	NaN
31801	ana	2026-01-19 15:50:00+00:00	2026-01-19 16:40:00+00:00	False	500	50	Travel and Transportation, Business and Finance	NaN
31802	peso pluma	2026-01-19 15:50:00+00:00	2026-01-20 08:20:00+00:00	False	10000	100	Entertainment	NaN
31803	tessa thompson	2026-01-19 15:20:00+00:00	2026-01-19 16:30:00+00:00	False	1000	400	Entertainment, Beauty and Fashion	NaN

31804 rows x 8 columns

```
Shape of DataFrame: (31804, 8)
Rows: 31804
Columns: 8
```

```
----- Data types of each column:
Column names:      query      object
1 -> query         start_date  object
2 -> start_date    end_date   object
3 -> end_date      active      bool
4 -> active        search_volume int64
5 -> search_volume increase_percentage int64
6 -> increase_percentage categories  object
7 -> categories    trend_breakdown object
8 -> trend_breakdown dtype: object
-----
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 31804 entries, 0 to 31803
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   query                 31804 non-null  object
1   start_date            31804 non-null  object
2   end_date              30421 non-null  object
3   active                31804 non-null  bool
4   search_volume         31804 non-null  int64
5   increase_percentage   31804 non-null  int64
6   categories            31804 non-null  object
7   trend_breakdown       11921 non-null  object
dtypes: bool(1), int64(2), object(5)
memory usage: 1.7+ MB
```

Statistical description:

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
query	31804	19709	online lottery	24	NaN	NaN	NaN	NaN	NaN	NaN	NaN
start_date	31804	8600	2026-03-19 08:40:00+00:00	136	NaN	NaN	NaN	NaN	NaN	NaN	NaN
end_date	30421	8755	2026-03-19 10:00:00+00:00	140	NaN	NaN	NaN	NaN	NaN	NaN	NaN
active	31804	2	False	30421	NaN	NaN	NaN	NaN	NaN	NaN	NaN
search_volume	31804.0	NaN	NaN	NaN	27489.891209	219937.68486	100.0	500.0	2000.0	10000.0	10000000.0
increase_percentage	31804.0	NaN	NaN	NaN	237.552666	256.353749	50.0	100.0	100.0	300.0	1000.0
categories	31804	245	Sports	15484	NaN	NaN	NaN	NaN	NaN	NaN	NaN
trend_breakdown	11921	11255	dallas stars	9	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Series:

	query
0	shedeur sanders
1	whoopi goldberg
2	mo williams
3	sloane stephens
4	canceled tv shows 2026

dtype: object

Type: <class 'pandas.core.series.Series'>

Selected column: query

	query
0	shedeur sanders
1	whoopi goldberg
2	mo williams
3	sloane stephens
4	canceled tv shows 2026

Selected columns: ['query', 'start_date', 'end_date']

	query	start_date	end_date
0	shedeur sanders	2026-03-30 18:40:00+00:00	NaN
1	whoopi goldberg	2026-03-30 18:40:00+00:00	NaN
2	mo williams	2026-03-30 18:00:00+00:00	NaN
3	sloane stephens	2026-03-30 17:50:00+00:00	NaN
4	canceled tv shows 2026	2026-03-30 17:50:00+00:00	NaN

Filtering condition applied on: query
Filtered rows: 31804

	query	start_date	end_date	active	search_volume	increase_percentage	categories	trend_breakdown
0	shedeur sanders	2026-03-30 18:40:00+00:00	NaN	True	200	50	Sports	NaN
1	whoopi goldberg	2026-03-30 18:40:00+00:00	NaN	True	5000	50	Politics	america first award, mike johnson
2	mo williams	2026-03-30 18:00:00+00:00	NaN	True	5000	300	Sports	mason williams basketball, mason williams
3	sloane stephens	2026-03-30 17:50:00+00:00	NaN	True	1000	200	Sports, Health	NaN
4	canceled tv shows 2026	2026-03-30 17:50:00+00:00	NaN	True	5000	200	Entertainment	renewed and cancelled tv shows 2026

Missing values in each column:

0	
query	0
start_date	0
end_date	1383
active	0
search_volume	0
increase_percentage	0
categories	0
trend_breakdown	19883

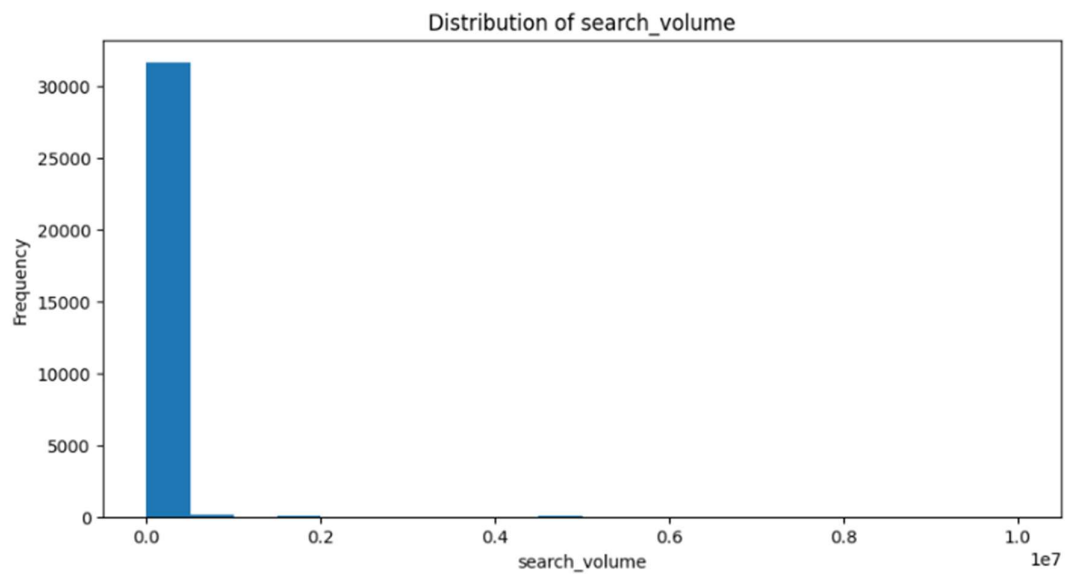
dtype: int64

Total missing values: 21266

Number of duplicate records: 0

	query
	online lottery
24	
	gma deals and steals
16	
	carlos alcaraz
16	
	vix
15	
	jessica pegula
15	
	rui hachimura
15	
	jake laravia
15	
	coco gauff
14	
	artificial intelligence news
14	
	truck accident lawyer
14	

dtype: int64



Result: